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Master's Thesis

Something with XAI stuff and making it actually work

Master's Program in Computer Science

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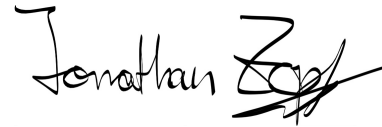
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Görlitz, July 7, 2026

Declaration of Originality

I hereby declare that I have written this thesis independently and have not used any sources other than those indicated. All passages that are taken verbatim or in spirit from published or unpublished sources are clearly marked as such. This work has not been submitted to any other examination board in the same or similar form.

Görlitz, July 7, 2026

A handwritten signature in black ink, appearing to read "Jonathan Zepf". The signature is stylized with a large, sweeping 'Z' and a final flourish.

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Chapter 1

Introduction

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Chapter 2

Background

2.1 Introduction to LLM Explainability

2.1.1 Definition and Purpose

Large language models (LLMs) have seen increasing adoption in recent years for tasks such as question answering [3], writing [19], and software development [44], but also in high-stakes domains such as healthcare [24, 17, 14]. However, since LLMs are complex deep learning models with millions of neurons and billions of parameters, they are often considered black boxes [32]. This makes it challenging to comprehend and evaluate their decisions.

Explainable AI (XAI) is a research field that seeks to make artificial intelligence algorithms transparent and understandable [13], thereby making the inherent black box less opaque. Central to this research field is the concept of explainability, whose precise definition remains disputed [35]. The term is described in the relevant ISO standard as the “level of understanding of how the AI-based system came up with a given result” [15]. Explainability aims to justify decisions and demonstrate that they are correct and unbiased in order to build trust with users [2].

The term *interpretability* is often used in a similar context, although there is a subtle distinction between the two concepts. Interpretability concerns understanding the inner workings of an AI model, whereas explainability focuses on making the decision process retraceable [9, 2].

2.1.2 Transparency as Regulatory Requirement

The recent increase in the importance of XAI research has also been driven by new regulatory measures. While multiple countries have introduced their own legislative frameworks to govern the use of AI, such as China [47] and the United States [5], the European Union’s AI Act [33] can be regarded as particularly relevant to explainability. Under this regulation, individuals affected by high-risk AI decisions have the right to obtain clear and meaningful explanations regarding the AI system’s role and the main contributing factors to the decision. Furthermore, high-risk AI systems must provide transparency and allow for human oversight [33]. The *Right to Explanation*

derived from the General Data Protection Regulation (GDPR) is also relevant to AI systems within the European Union [28].

2.1.3 Audience Groups

Providing explanations for reasoning processes or decisions made by an AI system can increase trust in the system among end users and experts [42, 10], while also assisting researchers and developers in debugging and fine-tuning the model [39]. Moreover, as argued in Chapter 2.1.2, explainability can be required by regulations and auditors, thereby forming another stakeholder group for XAI systems [10, 39].

It has also been noted that different audiences perceive and consume explanations differently [11, 10]. For instance, developers may expect technical information about the inner workings of the system, whereas end users may benefit more from concise and understandable explanations in natural language. Adapting explanations generated by an AI system to the expectations of the recipient has therefore become a subject of recent research [12, 26].

2.1.4 Aspects and Dimensions

Herrera [10] points out that, whereas traditional AI models usually follow fixed deterministic rules when generating explanations, LLMs complicate this process due to their more complex and advanced capabilities. The author therefore proposes four dimensions of explainability for large language models:

- **Faithfulness:** Faithfulness describes whether the explanation corresponds to the internal reasoning of the model [16].
- **Truthfulness:** Truthfulness describes whether the explanation conforms to real-world facts derived from an external authoritative reference. A truthful explanation does not contain factual errors, omissions, or distortions [10].
- **Plausibility:** Plausibility describes the degree to which an explanation is convincing and appears reasonable to humans [46, 16].
- **Contrastivity:** Contrastivity measures how well the explanation highlights differences between competing pieces of information. In classification tasks, an XAI system with high contrastivity clearly outlines *why* it selected one class over another [10, 27, 37].

It should be noted that there is still no broad scientific consensus within the XAI community regarding which aspects are essential for an explanation to be considered good, since descriptions, terminology, and definitions of these factors vary considerably [31, 2, 20, 27]. This issue is further amplified by regulatory and standardization frameworks, which remain comparatively vague in this regard [33, 15]. Nevertheless, Herrera’s [10] desiderata of faithfulness and plausibility in particular are supported by other literature, including [16, 46, 40]. Other authors, such as Lopes et al. [20], refer to faithfulness as *fidelity* and divide plausibility into several human-centered sub-aspects.

2.1.5 LLM Explainability Challenges

According to Zhao et al. [45, 22], explainability research for LLMs is difficult because understanding their internal processes becomes infeasible due to their enormous size and complex structure. Traditional methods such as gradient-based attribution [38] and SHAP values [21] are impractical for state-of-the-art LLMs with billions of parameters. This complexity also hinders in-depth analysis, debugging, and diagnosis of these models. Furthermore, the opaque black-box nature of LLMs amplifies this problem [32]. Many mainstream commercial models are only accessible via APIs [25], meaning that their model weights are not publicly available to users or researchers.

Despite these limitations, several technical approaches have been developed to gain insight into the black box and retrace important reasoning pathways within an LLM. These approaches cannot be discussed in detail within the scope of this thesis, but overviews of the most prominent methods are provided in [45, 22].

At the same time, since LLMs are specifically designed for natural language processing tasks, they also provide an opportunity to generate explanations directly in natural human language [45]. Most notably, this includes *ad-hoc* explanations where the model generates a justification for its decision after giving the prediction and *Chain-of-Thought* (CoT) explanations, in which LLMs reconstruct their own reasoning process. Prompting an LLM to generate CoT explanations has furthermore been shown to improve performance on complex reasoning tasks [43].

2.2 Faithfulness of LLM Explanations

2.2.1 Definition and Significance

Faithfulness describes the degree to which the explanation generated by an AI-powered system is grounded in the model’s actual internal reasoning process [10, 16, 34]. A faithful explanation should therefore reflect the model’s true decision path rather than merely providing a plausible post-hoc rationalization [10].

LLMs have the ability to give those explanations in natural language [43]. But unlike other explainability methods, the faithfulness of these are not guaranteed [10, 16]. In particular, an LLM may generate a natural language explanation that sounds convincing to users (high plausibility) while being entirely fabricated and not grounded in the model’s actual reasoning process (low faithfulness). Such scenarios can pose significant risks and lead to misplaced trust in these systems [1, 23, 10].

In particular, research by Turpin et al. [40] demonstrated that LLMs can systematically conceal internalized biases while providing plausible yet unfaithful justifications. When deliberately influenced to rely on prejudicial reasoning in their experiments, the models often used social biases such as gender or race as primary decision factors, but rarely mentioned them explicitly in their explanations. Instead, they resorted to alternative justifications based on conduct or circumstances. This behavior is especially concerning in applications such as recidivism prediction, where fairness may be undermined [8, 16]. Consequently, it is crucial not only to verify

whether an explanation appears reasonable, but also whether it accurately reflects the actual reasoning process of the model, particularly in high-stakes domains.

2.2.2 Estimation Approaches

General Challenges

As discussed in Section 2.1.5, the complex and opaque nature of large language models poses unique challenges for explainability research. These challenges are particularly pronounced when assessing the faithfulness of CoT explanations. Whereas plausibility can be evaluated by human judges [10], faithfulness is inherently tied to the model’s internal reasoning process [29, 16].

Faithfulness vs Self-Consistency

Some have argued such as [29] that faithfulness can not be measured and only self-consistency can be. Present here the case, why we still talk about measuring or estimating faithfulness.

Early Faithfulness Work

One of the earliest works addressing the estimation of faithfulness in LLMs was conducted by Turpin et al. [40]. In addition to demonstrating that CoT explanations can systematically omit contributing decision factors, the authors proposed a quantification approach for unfaithfulness on the BBH dataset. Their method measures the drop in model accuracy when biasing features are added to inputs that the model fails to mention in its explanations. However, the approach is labor-intensive, as CoT explanations generated by the LLM must be manually inspected by human annotators to determine whether the biased feature is mentioned. According to the authors, the method only evaluates faithfulness with respect to relatively small input modifications, whereas a practically useful metric should also be able to investigate reasoning behavior over larger input changes.

Another early contribution was presented by Chen et al. [6], who introduced the concept of *counterfactual simulatability*. This concept describes the degree to which humans can infer “the model’s outputs on diverse counterfactuals of the explained input.” In other words, after reading an explanation to an initial question, a human should be able to predict how the model will answer closely related counterfactual yes-or-no questions. Based on this concept, the authors propose two metrics: *simulation generality*, which measures the diversity of counterfactuals relevant to the explanation, and *simulation precision*, which measures the extent to which human expectations align with the model’s actual outputs on those counterfactuals.

Their evaluation pipeline first selects a question from a dataset and prompts the LLM to generate both an answer and an explanation. This explanation is subsequently used to generate corresponding counterfactual questions, a task performed by another LLM in order to scale to larger datasets. Human evaluators then construct a mental model based on the original explanation and state their expectations

regarding the answers to the counterfactual questions. Finally, the counterfactuals are presented to the LLM and the generality and precision scores are computed. This approach is notable because it supports both ad-hoc explanations and CoT explanations. It also represents an early example of using auxiliary LLMs to generate counterfactuals, thereby inheriting the limitations associated with LLM-as-a-Judge approaches [25].

Counterfactual Edits

Another influential use of AI-generated counterfactuals was introduced by Atanasova et al. [4]. The authors laid the foundation for a line of work commonly referred to as *counterfactual edits* [29, 25], which has subsequently been adopted and extended by later metrics [36, 25]. In these approaches, the LLM’s response to a counterfactual input serves as the basis for estimating faithfulness.

For their method, Atanasova et al. trained an auxiliary model to insert words into the original input, thereby generating counterfactuals. Both the original input and the modified version are then provided to the evaluated LLM. The resulting predictions and the explanation for the counterfactual input are subsequently analyzed. If the prediction changes due to the inserted words and those words are not mentioned in the explanation, the explanation is considered unfaithful. A major advantage of this approach is that it does not require human annotators [29].

Nevertheless, several limitations remain. The inserted modifications may shift the model’s attention to unrelated aspects of the input, meaning that the prediction change is not necessarily caused by the inserted feature itself [4]. Siegel et al. [36] additionally identify two practical shortcomings of this implementation of counterfactual edits. First, the method does not investigate whether highly influential features are more likely to be mentioned than weakly influential ones. In the extreme case, an AI system could theoretically achieve perfect faithfulness simply by repeating the entire input text. Second, the binary impact measurement only evaluates whether the prediction probability crosses the 50% threshold, thereby ignoring potentially meaningful probability shifts that remain below this boundary.

Correlation-Based Metrics

To address these limitations, Siegel et al. [36] proposed the Correlational Counterfactual Test (CCT), which significantly extends the approach of Atanasova et al. [4]. At its core, CCT measures faithfulness as the correlation between intervention impact scores and explanation mention scores [25].

First, the LLM receives an unmodified question from a dataset. The input is then perturbed and presented to the LLM again, producing prediction probability distributions for both the original and counterfactual versions. The counterfactual generation procedure largely follows the earlier work by Atanasova et al., although an auxiliary LLM is additionally used to filter nonsensical interventions. The Total Variation Distance (TVD) between the original and perturbed prediction distributions is then calculated in order to quantify the impact of the intervention. Subsequently, the

generated explanation is analyzed using substring matching to determine whether the intervention is explicitly mentioned, resulting in a binary explanation score. This procedure is repeated across multiple dataset examples, after which the Pearson correlation coefficient between intervention impacts and explanation mentions is computed to obtain the final CCT score.

Since the metric builds directly upon the approach by Atanasova et al. [4], several limitations are inherited, particularly with respect to counterfactual generation. For example, the counterfactuals are restricted to the insertion of individual adjectives or adverbs, limiting their semantic diversity. Furthermore, the metric does not account for cases in which an explanation refers to a synonym or semantically equivalent concept rather than the exact inserted word.

Output Contribution Metrics

Parcalabescu and Frank [29] argue that many existing metrics measure self-consistency of outputs rather than genuine faithfulness. They therefore propose their own self-consistency metric, CC-SHAP, which does not depend on modifying the input. Their approach compares the degree to which input tokens contribute to both the generated answer and the explanation. By measuring the convergence between these contributions using Shapley values, the method produces a continuous score.

This approach offers several advantages. It avoids the need for input perturbations, does not require semantic evaluation of explanations, supports both CoT and ad-hoc explanations, reveals patterns of consistency and inconsistency at the token level, and does not require auxiliary AI models during evaluation. However, these benefits come at the cost of considerable computational overhead, which may exceed that of other approaches [29].

CoT Corruption Methods

Lanham et al. [18] investigated faithfulness by deliberately corrupting CoT reasoning in order to test hypotheses regarding CoT behavior. First, the model is prompted to reason step-by-step about a multiple-choice problem. Given the original CoT, the LLM then produces a final answer. The CoT is subsequently modified through several interventions, including truncation, the insertion of mistakes using an auxiliary model, paraphrasing of earlier reasoning steps followed by regeneration of the remainder, and the insertion of filler tokens. The model is then asked again to provide an answer based on the modified CoT.

If the prediction remains unchanged despite the intervention, the explanation is considered unfaithful. Since the metric primarily serves as a tool for evaluating research hypotheses about CoT reasoning, it remains unclear how directly comparable it is to other faithfulness metrics. Parcalabescu and Frank [29] further argue that this approach assumes that the model fundamentally relies on the CoT in order to produce the correct prediction. However, experiments conducted by Lanham et al. [18] themselves suggest that CoT explanations only marginally improve prediction accuracy compared to models without CoT reasoning [29], thereby raising questions

regarding the practical usefulness of this method.

Faithfulness Through Unlearning

A final approach is presented by Tutek et al. [41], who propose a novel method called *Faithfulness by Unlearning Reasoning Steps* (FUR). The central idea is that if a reasoning step is genuinely used by the model to derive its answer, then removing the corresponding information from the model’s parameters should alter its prediction.

For a given LLM M and a test question, the model first generates a CoT explanation. The explanation is then decomposed into individual reasoning steps. For each step, the corresponding knowledge is selectively unlearned from the model parameters, producing a fine-tuned model M' that is discouraged from predicting the removed reasoning step in the given context. Subsequently, both the original model M and the modified model M' are asked the same question again, this time without requesting a CoT explanation, and their predictions are compared.

For the proposed ff-hard metric, which evaluates the faithfulness of the entire CoT, the result is binary. A score of 1 indicates that the models disagree on the most likely answer, suggesting high faithfulness, whereas a score of 0 indicates that the prediction remains unchanged and the explanation likely does not reflect the true reasoning process. One major strength of this approach is that it avoids several pitfalls associated with context perturbation methods, such as the model reconstructing the modified context. However, the authors themselves note that high recall cannot be guaranteed: if unlearning a reasoning step does not alter the prediction, this may indicate failed unlearning, the existence of multiple reasoning paths, or the possibility that the step was not actually used. Furthermore, the FUR approach is not applicable to closed-source API-only models.

2.3 Causal Concept Faithfulness Metric

2.3.1 Introduction

Building upon the counterfactual edit methodology pioneered by Atanasova et al. [4] and Siegel et al. [36], Matton et al. [25] proposed an updated evaluation metric in their work titled *Walk the Talk? Measuring the Faithfulness of Large Language Model Explanations*. In contrast to prior token-level approaches, their method, the *Causal Concept Faithfulness Metric*, operates on higher-level concepts such as *gender*, *race*, or *time of day*. Beyond producing a quantitative score, the metric also exposes semantic patterns linked to unfaithful behavior.

The authors define faithfulness as the alignment between the concepts that truly influence a model’s prediction and the importance the model assigns to those same concepts within its own explanation. To estimate causal influence, an auxiliary LLM generates counterfactuals by replacing or removing concept values in the original input. Following the correlational approach of Siegel et al. [36], prediction distributions are then collected for both original and counterfactual questions to estimate the true causal effect. Subsequently, another helper LLM extracts the concepts identified as

important in the explanation, enabling the calculation of the explanation-implied effect. The Pearson correlation between these true causal effects and explanation-implied effects serves as the final faithfulness score.

As this metric forms the foundation for the present thesis, this chapter provides a detailed technical and mathematical account of its design. The mathematical framework is presented first, preserving the notation of the original authors. This is followed by a walkthrough of the evaluation pipeline that produces the final metric. The chapter concludes by discussing the promising potential of the *Causal Concept Faithfulness Metric* for quantifying the faithfulness of LLM-generated explanations, while also examining its existing limitations. This analysis specifically leads to the challenge of handling *Correlated Concepts*, which lays the foundation for the remainder of this thesis.

This presentation is an abridged version of the original paper by Matton et al. [25] and serves only as an introduction. The interested reader is referred to the original work for further details.

2.3.2 Mathematical Foundation

Problem Setting

The goal is to evaluate the faithfulness of explanations generated by an opaque large language model under evaluation, denoted by M , for a dataset of questions $X = \{x_1, \dots, x_N\}$. Matton et al. formulate their metric for so-called *context-based questions*, where each response r produced by the model consists of both a selected answer $y \in Y$ from a predefined set of answer options and a natural language explanation e , i.e.,

$$r = (y, e).$$

The underlying assumption is that the explanation implicitly or explicitly indicates which parts of the input influenced the model’s prediction. Rather than considering these parts as individual tokens or words, Matton et al. model them as high-level *concepts*. Consequently, the proposed metric measures the discrepancy between “the set of concepts that LLM explanations imply are influential and the set that truly are” [25].

Concepts and Categories

For each question x , a set of concepts

$$C = \{C_1, \dots, C_M\}$$

is identified. Unlike token-based approaches, concepts correspond to semantically meaningful properties of the input rather than low-level textual features. Each concept C_m can assume different possible values $c_m \in \mathbb{C}_m$, where $\mathbb{C}_m$ denotes the domain of all plausible values for concept C_m .

A key assumption of the metric—which will later be challenged in this thesis—is that concepts are “disentangled”. In other words, each concept can be modified

independently without affecting the remaining concepts. This assumption allows causal interventions to be applied to individual concepts while keeping all others fixed.

Furthermore, every concept is assigned to a higher-level category from the set

$$K = \{K_1, \dots, K_L\},$$

for example *gender*, *ethnicity*, or *medical symptoms*. Besides improving the interpretability of the resulting analysis, these categories are also required for the Bayesian hierarchical model employed by Matton et al. to estimate causal concept effects through partial pooling.

Causal Concept Effects (CE)

To estimate the actual influence of a concept on the model’s prediction, Matton et al. generate counterfactual questions. A counterfactual,

$$x_{c_m \rightarrow c'_m},$$

is obtained by replacing the original value c_m of concept C_m with an alternative value c'_m , while leaving the remainder of the input unchanged. Let

$$\mathbb{C}'_m = \mathbb{C}_m \setminus \{c_m\}$$

denote the set of all alternative values of concept C_m .

The causal concept effect is then defined as the average Kullback–Leibler divergence between the model’s prediction distribution for the original question and each corresponding counterfactual:

$$CE(x, C_m) = \frac{1}{|\mathbb{C}'_m|} \sum_{c'_m \in \mathbb{C}'_m} D_{\text{KL}} \left(\mathbb{P}_M(Y \mid x_{c_m \rightarrow c'_m}) \parallel \mathbb{P}_M(Y \mid x) \right). \quad (2.1)$$

Intuitively, concepts whose modification substantially changes the model’s output receive a high causal concept effect, whereas concepts with little influence receive values close to zero.

Explanation-Implied Effects (EE)

While the causal concept effect measures the actual influence of a concept on the model’s prediction, the explanation-implied effect estimates how influential the model itself claims the concept to be within its explanation. If the explanations are faithful, concepts with large causal effects should be mentioned more frequently than concepts with only a negligible influence.

Matton et al. therefore define the explanation-implied effect as the probability that the model’s explanations for both the original question and its counterfactuals identify concept C_m as being causally relevant:

$$EE(x, C_m) = \frac{1}{|\mathbb{C}_m|} \sum_{c'_m \in \mathbb{C}_m} \mathbb{P}_M(C_m \in E \mid x_{c_m \rightarrow c'_m}). \quad (2.2)$$

In practice, this quantity is estimated by automatically extracting the concepts referenced in the generated explanations using an auxiliary LLM.

2.3.3 Causal Concept Faithfulness

After computing both the causal concept effects and the explanation-implied effects for every concept of a question, the final faithfulness score can be determined. Specifically, the Pearson correlation coefficient between the vectors of causal concept effects and explanation-implied effects is calculated:

$$F(x) = \text{PCC}(CE(x, C), EE(x, C)). \quad (2.3)$$

This yields a result within $[-1, 1]$. A high positive correlation indicates that concepts which genuinely influence the model’s prediction are also consistently identified as influential within the explanation, whereas a low or even negative correlation indicates unfaithfulness.

Finally, the dataset-level faithfulness of model M is obtained by averaging the question-level faithfulness scores over all questions in the dataset:

$$F(X) = \frac{1}{|X|} \sum_{x \in X} F(x). \quad (2.4)$$

2.3.4 Evaluation Pipeline

Extracting Concepts with Values and Categories

With a given question x drawn from the dataset X the concept set C has to be determined of all concepts that are included in the question. Since manual annotation by human reviewers would be tedious and impractical for an automated evaluation, an auxiliary LLM A is tasked with this work. It should be noted that A does not necessarily have to be the same model as the LLM-under-test M . The prompt instructs A to first identify the concepts C for the given question, as well as determine their respective category from the dataset-wide set K (however a list of possible categories is not supplied to the auxiliary llm). After that, in a second prompt, the LLM is then instructed to extract the current value of each identified concept as well as a plausible alternative value which results in $\mathbb{C}_m$ for every $C_m \in C$. All prompts within this pipeline feature examples to facilitate the LLM sticking to the desired output format. The result is then parsed and saved by the evaluation program and it can be continued to proceed to generate actual counterfactuals.

Generating counterfactual questions

By identifying the concepts involved in a question with their respective values, the cornerstone has been set for creating counterfactual question $x_{c_m \rightarrow c'_m}$. More specifically, to do so, the concept C_m , its current value c_m and the alternative value c'_m are passed down and A is instructed to overhaul x to $x_{c_m \rightarrow c'_m}$ while keeping all other concepts unchanged. So in summary, the auxiliary LLM A is queued to rephrase x

with a new concept value. (with the least possible change to the structure or words not skew results, but the question should still be coherent).

Besides creating counterfactuals by changing the value of a concept C_m it might be, depending on the experimental setup and dataset, instead desirable to instead remove and obfuscate a concepts from a question. For doing so A is instructed to make C_m undeterminable from question x .

Collecting model responses

In order to be able to calculate the Causal Concept Effects (CE), its required to have access to the probability distributions of M 's choices for both original and all counterfactual questions $x_{c_m \rightarrow c'_m} : c'_m \in \mathbb{C}'m$. To reconstruct those probability distributions a certain number of samples is being collected for every version of x by repeatedly prompting the LLM-under-test M . By doing so, not only are the answer choices obtained but also the explanations collected, which shall be subject to further analysis as well.

Estimating Explanation-Implied Effects

In order to get the Explanation-Implied Effects its needed to analyze the explanations on for looking if M indicates that C_m has a causal effect and might shift its prediction with alternative values. For automation, this step is also done by the auxillary LLM A . However: the auxillary LLM is instructed that one pitfall here is to assume that all concepts which are mentioned are influential and important. Instead A in method of Matton et al. is instructed to look for the context those concepts are mentioned in the explanation and not for there mere presence. (This is very awkwardly written, need to rework that).

After analyzing all samples for all versions (original and counterfactuals) of a question x a DataFrame can be compiled for all $C_m \in x$ with the respective probability (averaged across all analyzed samples) that C_m is explained to have a causal effect. By using those probabilities its no difficulty to calculate the explanation-implied effects for the entire question (see equation 2.2).

Estimation Causal Concept Effects

Given that the model responses where already collected, it would be trivial to calculate the Kullback-Leibler divergence between the distribution of responses pre- and post-intervention for a given concept. However, Matton et al. point out that this approach can be inaccurate if only a small number of responses have been collected. By assuming that "that similar concepts have a similar magnitude of effect on LLM M 's answers within the context of a dataset" [25] they instead introduce a Bayesian hierarchical model within the context of the entire dataset. This allows the evaluation pipeline to "draw" information on the magnitude of concept effects of similar concepts across questions ($x \in X$). This multinomial logistic regression-approach is too rich and complex to be discussed in mathematical detail within this thesis, especially since it has only limited pertinence to the problem that should be addressed.

By having made the calculation of the KL-divergence between both answer distributions more stable, it's now possible to proceed to calculate the causal concept effects across all counterfactual values (see 2.1) for a given concept and question.

2.3.5 Estimating Causal Concept Faithfulness

This is the last step within Matton et al.'s' evaluation pipeline. The Causal Concept faithfulness has been defined as the Pearson correlation coefficient between the Causal Concept Effects and the Explanation-Implied Effects. However just with the CE-vector calculation the authors see a similar problem: for most questions the number of concepts involved is small, leading to only a handful of datapoints, and will make the results unreliable.

To solve this obstacle, Matton et al. suggest again to employ a Bayesian hierarchical model on the dataset-level, which "draws" data on faithfulness from other questions. They assume that faithfulness scores for questions within a dataset are rather similar due to a high degree of similarity of content. To make this linearly regression possible, the X and Y datapoints are prepared by normalizing their standard deviation to the same value. After that, a global regression parameter of the expected PCC is defined, serving as a prior on question-level regression coefficients. After fitting the model the posteriors of the question-regression coefficients are used as the question-level faithfulness. Extracting the global-regression coefficient would yield the $F(x)$ -value of the dataset.

2.3.6 The Case for this Metric

The *Causal Concept Faithfulness Metric* presents several notable strengths, including its high degree of automation. Its black-box treatment of the LLM under test makes it applicable to virtually any model, regardless of internal architecture or API-only access. Furthermore, the high-level concept-based approach aligns more closely with real-world applications and the semantic reasoning of LLMs than do token-level methods such as that of Atanasova et al. [4]. Matton et al. [25] also emphasize that their metric identifies patterns of unfaithfulness, providing insight into how explanations mislead and supporting informed decisions about model deployment in given environments. This potential alone makes the metric a highly promising foundation, warranting the investment of resources to address its remaining weaknesses.

For instance, many limitations stem from its reliance on LLM-based automation. Matton et al. [25] acknowledge that even a capable proprietary auxiliary model, GPT-4o, at times introduced errors into the pipeline. In concept extraction, all identified concepts were present in the question, but not all satisfied the disentanglement constraint. Counterfactual generation also exhibited some deficits, including the unintended modification of non-target concepts and the inconsistent replacement of a concept mentioned in multiple locations. Such errors can propagate through subsequent stages and skew the results even though their occurrence is considered rare. More generally, the need to adapt auxiliary LLM prompts to specific datasets was identified as a practical disadvantage by the authors themselves.

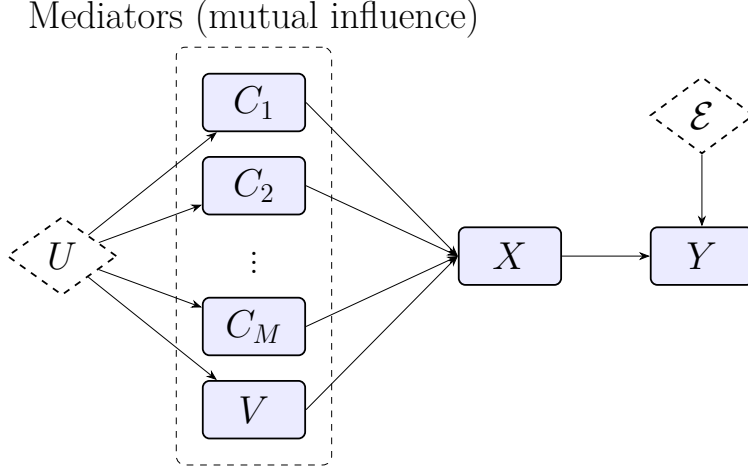


Figure 2.1: The DGP for original dataset-questions and the responses according to Matton et al. [25]

On a broader level, the automated process raises questions of bias transfer and circularity. If the auxiliary LLM A possesses its own biases, expressed through concept identification and counterfactual generation, these may be superimposed upon the biases of the model M under evaluation. Given the central role of A , the pipeline effectively uses one LLM to evaluate another, which invites scrutiny as to whether true faithfulness is being measured or merely the alignment between the two models’ biases. Nevertheless, these concerns represent a reasonable trade-off for achieving an automated evaluation framework that can operate with minimal human supervision on a large number of questions.

2.4 The Correlated Concepts Problem

2.4.1 Assumptions in the observational graph

In part B of the appendix within *Walk the Talk? Measuring the Faithfulness of Large Language Model Explanations* Matton et al. [25] discuss their reasoning behind their definition of causal effects that concepts have. They use a causal graph to discuss their reasoning behind the data generation process for a given question. With $C_m M; m = 1$ being the concepts in the question acting as mediating variables between possible questions X and the state of the world as an unobserved variable U . V is also a mediating variable, that accounts for other aspects of X not covered by the concepts $C_m M; m = 1$ such as writing style within a question. Y is the answer the LLM makes while ϵ remains unobserved and represents the stochasticity of the model. ?? shows those nodes.

Matton allows that all mediating variables can influence each other. This means that the concepts $C_m M; m = 1$ can affect each other and can also act on the rest of the question text V . The authors also allow the confounder U to create correlations between them. To symbolize this layer of mutual influence and correlation, a dotted

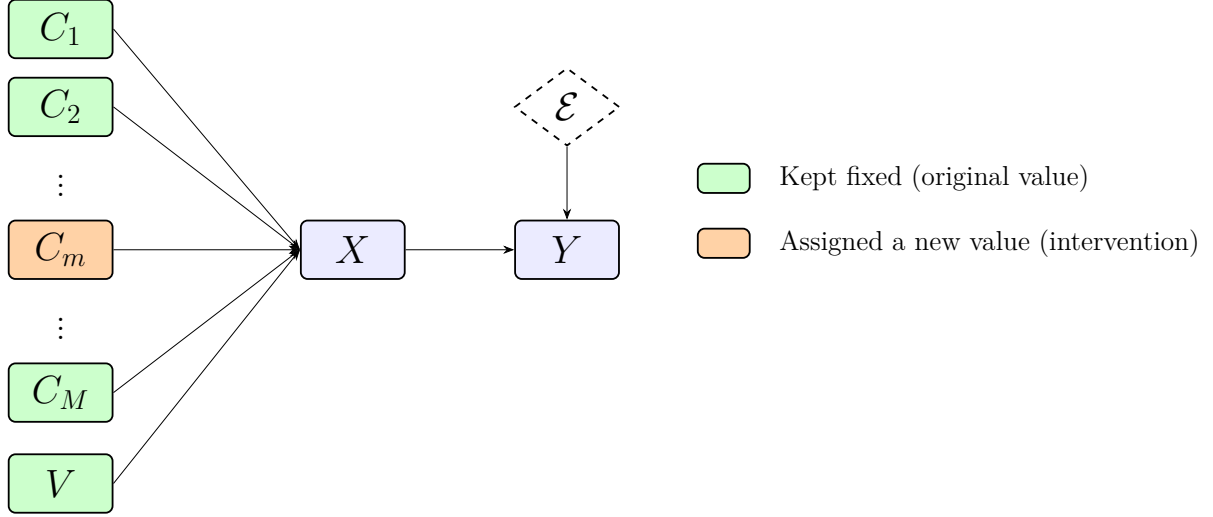


Figure 2.2: The Causal Graph where the value of concept C_m is changed and all other concepts and V are kept fixed

line is drawn around the mediators in ??.

On a practical level this means that the concept candidates’ genders might causally influence the concept candidates’ names (because gendered names are more likely), or that the unobserved “state of the world” U makes both age and experience level appear together in a question. In the observational data (the given questions), these variables are therefore correlated and not independent.

2.4.2 The “disentanglement” assumption for interventions

Matton et al. write that merely measuring the total causal effect a concept C_m has on the answer distribution Y would be undesired since the given explanation by the LLM is not expected to mention “*concepts that influenced other concepts that then influenced its answer*” but instead supposed to name the concepts that directly influenced its answer [25]. So in order to bring the Concept Implied Effect and Explanation Implied Effect properly into conjunction, the direct effect should be measured instead, where any effect mediated by other variables is being discarded [30] is measured.

In order to measure the direct effect, it’s necessary for one concept C_m to be modified while keeping all other concepts $C_{i \neq m} \text{ for all } i \dots M$ (fix the mathematical notation) as well as the rest of the question text V at their respective original values [30].

Pearl [30] calls this a surgical intervention: any mutual influences are severed between the mediating variables leading to a changed causal structure of the model where the intervened variable becomes independent of other causes [30, 7]. In order to measure the direct effect, Matton et al. therefore have to assume that concepts are entirely disentangled - that they one can be changed without modifying any other concepts or even V .

Allowing such surgical interventions is no contradiction to the assumption in 2.4.1

that mutual influences and observational correlations between the mediating variables can exist. Constructing models that permit interventions on individual mediating variables does not require those variables to be statistically independent in observational data. [30].

The graph ?? illustrates the effects of a surgical concept intervention on the causal graph. Fixed and the changed value C_m are indicated by colors to show what was changed and what kept. Note that the confounder U is removed since it does not affect the fixed mediators anymore. Furthermore due to the fixure of values for V and $C_i M; i = 1 \text{ and } i = m$ (so all concepts that are not the one to be changed) any dependencies and mutual influences within the mediators cease to exist, indicated by the removal of the dotted border around them.

2.4.3 The practical limit

While as argued before there is no theoretical problem in assuming concepts are disentangled and surgical interventions are possible several practical aspects must be considered. First it should be considered idealistic to assume that the rest of the text V can be kept fixed in any arbitrary intervention. There might be examples where the style of writing might appear unnatural or incoherent when it's kept fixed with a changing concept value.

However most notably, in some instances concepts might be correlated within the training data set used by the LLM. Matton et al. themselves acknowledge that in such cases intervening on only one and keeping other correlated ones fixed will lead to unreliable results in calculating the direct effect [25]. They argue that this can be particularly pronounced with removal-based counterfactuals: the LLM might be able to reconstruct the removed concept value by looking at the unchanged concepts where the LLM expects them to be correlated with the removed one. For instance in a job application context, the applicants gender might be easily reconstructed from the first name in many cases. In summary in such cases the "surgical intervention" becomes not surgical anymore since it leaves traces in the question.

Matton et al. illustrate this issue with a question within the MedQA dataset. They note that the concepts *the patient's eating disorder* and *the patient's self-perception of weight* each have a suprisingly low CE-score. They hypothesize that this can be explained because of the LLM reconstructing the other easily when removing only one since both concepts can be said to correlate.

On a more general note this can lead to the causal effect to be underestimated for truly influential concepts. This might widen the gap between causal effect and explanation implied effect and might punish "intelligent" and "sneaky" LLMs with a bad faithfulness score leading to skewed results. Even though the presence of correlation varies significantly between questions and datasets, this constitutes a major flaw. Therefore finding a reliable fix for this limitation is important to promote the practical adoption of the *Causal Concept Faithfulness Metric*. With the problem clearly established, this is the basis for this thesis from now on: to construct, implement and then validate an extension of the metric by Matton et al. that considers correlated concepts in Causal Effect calculation.

Chapter 3

Implementation

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